

Lecture 1

Dr. Kiran Kumari



Department of Electrical Engineering
Indian Institute of Science
Bengaluru, India

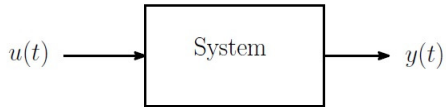
E1 248
Sliding mode control and its application

Prerequisite: Transfer function, Laplace transform, Solution of differential equations

Topics covered:

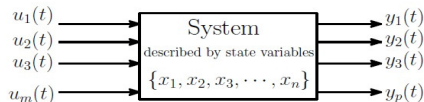
- What is state?
- Why do we need state variables?
- State space model for RLC circuit.
- Invariance of the transfer function for different representations.
- Different state space realization

- **Plant:** A plant may be a piece of equipment, perhaps just a set of machine parts functioning together, the purpose of which is to perform a particular operation.
- **System:** A system is a combination of components that act together and perform a certain objective. The concept of the system can be applied to abstract, dynamic phenomena.
- A system can be physical, biological, economic, etc.
- The mathematical model of a system is linear, if it obeys the principle of superposition and homogeneity.
- **Linear system:** A mathematical model is linear if the differential equation describing it has coefficients, which are either functions only of the independent variable or are constants.
- **Linear time-varying:** If coefficients of the describing differential equations are functions of time (the independent variables), then the mathematical model is linear time-varying.
- **Linear time-invariant:** On the other hand, if the coefficients of the describing differential equations are constants, the model is linear time-invariant.



Definition of state:

The state of a system is the set of variables, which, if known at time t_0 together with input $u(t)$ for $t \geq t_0$, the system behavior for all $t \geq t_0$ can be determined.



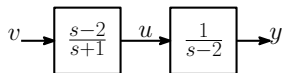
Definition of state:

The state of a system is the set of variables, which, if known at time t_0 together with input $u(t)$ for $t \geq t_0$, the system behavior for all $t \geq t_0$ can be determined.

Why we need state variables?

1. This approach can be applied to linear or nonlinear, time-variant or time-invariant systems.
2. Can handle SISO and MIMO systems with equal ease.
3. The internal behavior of the systems is complicated, and it can not be described by a transfer function.
4. The chosen state variables need not represent physical or measurable variables.
5. Matrix transformations can be used to convert from one state variable representation to the other.

Consider the following block diagram:



The transfer function $G(s) = \frac{Y(s)}{V(s)} = \frac{s-2}{s+1} \cdot \frac{1}{s-2} = \frac{1}{s+1}$.

- First calculate the step response:

$$v(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases} \implies V(s) = \frac{1}{s} \implies Y(s) = \frac{1}{s(s+1)}$$

- The output response is solved as $y(t) = \mathcal{L}^{-1} \left[\frac{1}{s} - \frac{1}{s+1} \right] = 1 - e^{-t}$, $t \geq 0$.
- Is this calculation of $y(t)$ represent true output of the system?
- It can be seen from the actual implementation of this system (on an analog computer) that the output y saturates or the system burns out.
- From the first transfer function, we get

$$\dot{u} = -u + \dot{v} - 2v \implies \dot{u} - \dot{v} = -(u - v) - 3v.$$

- From the second transfer function, we get

$$\dot{y} = 2y + u.$$

- By defining state variables $x_1 = u - v$ and $x_2 = y$, we get the following differential equations

$$\begin{aligned}\dot{x}_1 &= -x_1 - 3v & x_1(0) &= x_{10} \\ \dot{x}_2 &= x_1 + 2x_2 + v & x_2(0) &= x_{20}.\end{aligned}$$

- Solve the above differential equation using Laplace transforms:

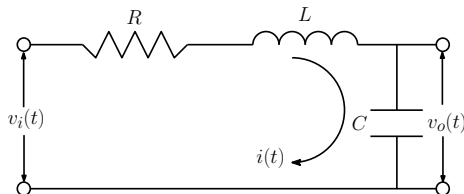
$$\begin{aligned}sX_1(s) - x_1(0) &= -X_1(s) - 3V(s) \\ sX_2(s) - x_2(0) &= X_1(s) + 2X_2(s) + V(s)\end{aligned}$$

- Calculate $Y(s) = X_2(s)$ using above equations as

$$Y(s) = \frac{x_2(0)}{s-2} + \frac{x_1(0)}{(s-2)(s+1)} + \frac{v(s)}{s+1}$$

- This gives $y(t) = e^{2t}x_{20} + \frac{1}{3}(e^{2t} - e^{-t})x_{10} + e^{-t}v(t)$.
- For non-zero initial conditions, the response of the system deviates as $t \rightarrow \infty$.

State space model for RLC circuit



- From Kirchoff's Laws:

$$L \frac{di}{dt} + Ri + v_o(t) = v_i(t)$$

$$\frac{dv_o}{dt} = \frac{1}{C} i(t)$$

- Using the above equations, one can write

$$LC \frac{d^2 v_o}{dt^2} + RC \frac{dv_o}{dt} + v_o(t) = v_i(t).$$

- **Step 1:** Express the highest derivative of $v_o(t)$ in terms of all other quantities

$$\frac{d^2 v_o}{dt^2} = -\frac{R}{L} \frac{dv_o}{dt} - \frac{1}{LC} v_o(t) + \frac{1}{LC} v_i(t)$$

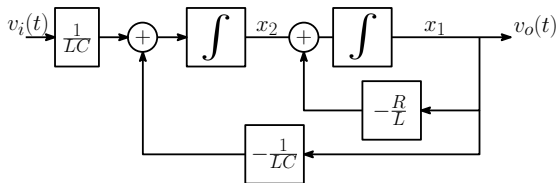
- **Step 2:** Integrate both sides $n = 2$ times

$$\iint \ddot{v}_o(t) = -\frac{R}{L} \iint \dot{v}_o(t) - \frac{1}{LC} \iint v_o(t) + \frac{1}{LC} \iint v_i(t)$$

- **Step 3:** Further reduce to represent above equation only in v_o and v_i

$$v_o(t) = -\frac{R}{L} \int v_o(t) - \frac{1}{LC} \iint v_o(t) + \frac{1}{LC} \iint v_i(t) \quad (1)$$

- **Step 4:** Draw $n = 2$ integrators. Put output (v_o) at the right end and input (v_i) at the left end.
- **Step 5:** Draw above block diagram according to (1).



- Label the output of the integrators as x_1, x_2 .
- x_1, x_2 state variables and they are related to v_o and v_i through a set of first-order differential equations.

State equations

$$\begin{aligned}\dot{x}_1(t) &= -\frac{R}{L}x_1(t) + x_2(t) \\ \dot{x}_2(t) &= -\frac{1}{LC}x_1(t) + \frac{1}{LC}v_i(t) \\ y(t) &= v_o(t) = x_1(t)\end{aligned}$$

State space model

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -\frac{R}{L} & 1 \\ -\frac{1}{LC} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{LC} \end{bmatrix} v_i$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

- A differential equation of order n can be represented as n first-order differential equations.

- It is possible to write the same differential equation in different state space representations.
- Let $x_1 = v_o$ and $x_2 = \frac{dv_o}{dt}$.
- Calculate $\dot{x}_2$ from $\frac{d^2 v_o}{dt^2} = -\frac{R}{L} \frac{dv_o}{dt} - \frac{1}{LC} v_o(t) + \frac{1}{LC} v_i(t)$.

State equations

$$\begin{aligned}\dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= -\frac{1}{LC} x_1(t) - \frac{R}{L} x_2(t) + \frac{1}{LC} v_i(t) \\ y(t) &= v_o(t) = x_1(t)\end{aligned}$$

State space model

$$\begin{aligned}\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ -\frac{1}{LC} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{LC} \end{bmatrix} v_i \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\end{aligned}$$

- Standard choice of states:
For a mechanical system: Displacement $x(t)$ and Velocity $v(t)$.
For an electrical system: Inductor current $i(t)$ and Capacitor voltage $v(t)$.

Invariance of transfer function

Laplace transform of (2) gives

$$sX(s) - x(0) = AX(s) + BU(s)$$

$$Y(s) = CX(s)$$

$$\Rightarrow (sI - A)X(s) = x(0) + BU(s) \text{ for } x(0) = 0$$

$$\Rightarrow X(s) = (sI - A)^{-1}BU(s)$$

$$\Rightarrow Y(s) = C(sI - A)^{-1}BU(s)$$

$$\Rightarrow G(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B.$$

Invariance of transfer function

Laplace transform of (2) gives

$$sX(s) - x(0) = AX(s) + BU(s)$$

$$Y(s) = CX(s)$$

$$\Rightarrow (sI - A)X(s) = x(0) + BU(s) \text{ for } x(0) = 0$$

$$\Rightarrow X(s) = (sI - A)^{-1}BU(s)$$

$$\Rightarrow Y(s) = C(sI - A)^{-1}BU(s)$$

$$\Rightarrow G(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B.$$

Let there exists a transformation matrix T such that $x = T\hat{x}$, so

$$\dot{\hat{x}} = T^{-1}\dot{x} = T^{-1}Ax + T^{-1}Bu$$

$$= T^{-1}AT\hat{x} + T^{-1}Bu$$

$$\dot{\hat{x}}(t) = \hat{A}\hat{x}(t) + \hat{B}u(t)$$

$$y(t) = CT\hat{x} = \hat{C}\hat{x}(t).$$

- For the RLC circuit example,

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} v_o(t) \\ \dot{v}_o(t) \end{bmatrix}, \quad \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} v_o(t) \\ \hat{x}_2 \end{bmatrix}, \quad \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{R}{L} & 1 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix}$$

- Transfer function of new representation:

$$\begin{aligned} \hat{G}(s) &= \hat{C}(sI - \hat{A})^{-1} \hat{B} \\ &= CT (sI - T^{-1}AT)^{-1} T^{-1}B \\ &= CT (sT^{-1}T - T^{-1}AT)^{-1} T^{-1}B \\ &= CT (T^{-1}(sI - A)T)^{-1} T^{-1}B \\ &= CT (T^{-1}(sI - A)^{-1}T) T^{-1}B \\ &= C(sI - A)^{-1}B = G(s). \end{aligned}$$

- Hence, the transfer function does not change for different representations.
- The new state space representation still represents the same system.

Input-Output Relations-Transfer Functions

- Let the state space description of a system be

$$\begin{aligned}\dot{X} &= AX + BU \\ Y &= CX\end{aligned}$$

The above equation describes the internal behavior of the system.

- The transfer function of this system is

$$\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B$$

The transfer function describes the external behavior of the system, which is the input-output behavior.

First Companion Form

- Consider the following transfer function

$$\frac{Y(s)}{U(s)} = \frac{1}{s^k + a_1 s^{k-1} + \dots + a_k}$$

which can be written as

$$(s^k + a_1 s^{k-1} + \dots + a_k) Y(s) = U(s) \quad (3)$$

- Differential equation corresponding to (3) is

$$\begin{aligned} D^k y + a_1 D^{k-1} y + \dots + a_k y &= u \\ \Rightarrow D^k y &= -a_1 D^{k-1} y - a_2 D^{k-2} y \dots - a_k y + u \end{aligned} \quad (4)$$

Assume states $x_1, x_2, \dots$ such that

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\cdot \quad \cdot \quad \cdot$$

$$\cdot \quad \cdot \quad \cdot$$

$$\dot{x}_{k-1} = x_k$$

Now from (4), following differential can be obtained

$$\dot{x}_k = -a_k x_1 - a_{k-1} x_2 - \dots - a_1 x_k + u$$

The matrices corresponding to the system

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdot & \cdot & 0 \\ 0 & 0 & 1 & \cdot & \cdot & 0 \\ 0 & 0 & 0 & 1 & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ -a_k & -a_{k-1} & \cdot & \cdot & \cdot & -a_1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \cdot \\ \cdot \\ \cdot \\ 1 \end{bmatrix},$$
$$C = \begin{bmatrix} 1 & 0 & 0 & \cdot & \cdot & 0 \end{bmatrix}$$

- Consider the more general transfer function

$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_0 s^k + b_1 s^{k-1} + \dots + b_k}{s^k + a_1 s^{k-1} + \dots + a_k} \quad (5)$$

- The development is aided by the introduction of an intermediate variable $Z(s)$ as follows

$$\frac{Y(s)}{U(s)} = \frac{Y(s)}{Z(s)} \cdot \frac{Z(s)}{U(s)} = \frac{b_0 s^k + b_1 s^{k-1} + \dots + b_k}{s^k + a_1 s^{k-1} + \dots + a_k}$$

- Identify the first factor with the numerator and the second factor with the denominator, i.e.

$$\frac{Y(s)}{Z(s)} = b_0 s^k + b_1 s^{k-1} + \dots + b_k \quad (6)$$

and

$$\frac{Z(s)}{U(s)} = \frac{1}{s^k + a_1 s^{k-1} + \dots + a_k} \quad (7)$$

Realization of the transfer function from u to z has already been developed.

- From (6)

$$Y(S) = (b_0 s^k + b_1 s^{k-1} + \dots + b_k) Z(S)$$

$$\Rightarrow y = b_0 D^k z + b_1 D^{k-1} z + \dots + b_k z$$

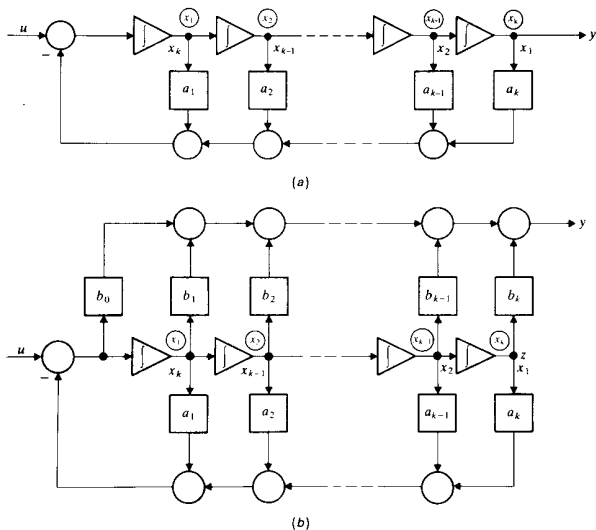


Figure: State-space realization of transfer functions in first companion form for

(a) $H(s) = \frac{1}{s^k + a_1 s^{k-1} + \dots + a_k}$ (b) $H(s) = \frac{b_0 s^k + b_1 s^{k-1} + \dots + b_k}{s^k + a_1 s^{k-1} + \dots + a_k}$

- The inputs to the integrators in the chain are the k successive derivatives of z , as shown in the above Figure.
- The state equations are the same as before. The output equation is found by careful examination of the block diagram.
- Note that there are two paths from the output of each integrator to the system output. One path upward through the box labeled b_i and a second path down through the box labeled a_i .
- As a consequence, the output equation is written as

$$y = (b_k - a_k b_0) x_1 + (b_{k-1} - a_{k-1} b_0) x_2 + \dots + (b_1 - a_1 b_0) x_k + b_0 u$$

Hence

$$C = \begin{bmatrix} b_k - a_k b_0 & b_{k-1} - a_{k-1} b_0 & \cdot & \cdot & \cdot & b_1 - a_1 b_0 \end{bmatrix}$$

$$D = [b_0]$$

- If the direct path through b_0 is absent then the matrix D is zero and the C matrix contains only the b_i coefficients.
- In the first canonical form realization, the input is connected directly to the first integrator in the chain and the output is a linear combination of the outputs of the integrator.

A variant of the structure of the previous Figure in which the output is taken directly from the last integrator, but the input is connected to all the integrators is shown in the figure below.

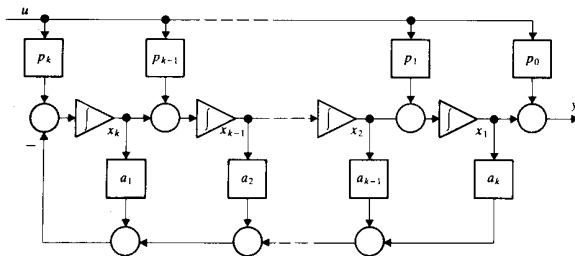


Figure: Alternative first companion form of realization of transfer function

$$H(s) = \frac{b_0 s^k + b_1 s^{k-1} + \dots + b_k}{s^k + a_1 s^{k-1} + \dots + a_k}$$

- The feed-forward gains $p_1, p_2, \dots, p_k$ are in general not equal to coefficients $b_1, b_2, \dots, b_k$ of transfer function, but must be obtained by solution of a set of linear algebraic equations, which may be derived as follows.

$$\begin{aligned}
 \dot{x}_1 &= x_2 + p_1 u \\
 \dot{x}_2 &= x_3 + p_2 u \\
 &\vdots \\
 &\vdots \\
 \dot{x}_{k-1} &= x_k + p_{k-1} u \\
 \dot{x}_k &= -a_1 x_k - \dots - a_k x_1 + p_k u
 \end{aligned} \tag{8}$$

and

$$y = x_1 + p_0 u \tag{9}$$

- Differentiating (9) k times and using (8), we obtain

$$\begin{aligned}
 Dy &= x_2 + p_1 u + p_0 Du \\
 D^2 y &= x_3 + p_2 u + p_1 Du + p_0 D^2 u \\
 &\vdots \\
 &\vdots
 \end{aligned}$$

$$D^{k-1} y = x_k + p_{k-1} u + p_{k-2} Du + \dots + p_1 D^{k-2} u + p_0 D^{k-1} u$$

- From (10) and (9) we thus get

$$\begin{aligned}
 D^k y + a_1 D^{k-1} y + \dots + a_{k-1} D y + a_k y &= (p_k + a_1 p_{k-1} + \dots + a_{k-1} p_1 + a_k p_0) u \\
 &\quad + (p_{k-1} + \dots + a_{k-2} p_1 + a_{k-1} p_0) D u \\
 &\quad + \dots + (p_1 + a_1 p_0) D^{k-1} u + p_0 D^k u
 \end{aligned}
 \tag{11}$$

- In order for (11) to represent the original transfer function, it is necessary that

$$\begin{aligned}
 p_0 &= b_0 \\
 p_1 + a_1 p_0 &= b_1 \\
 &\cdot \quad \cdot \quad \cdot \\
 &\cdot \quad \cdot \quad \cdot \\
 p_{k-1} + \dots + a_{k-2} p_1 + a_{k-1} p_0 &= b_{k-1} \\
 p_k + a_1 p_{k-1} + \dots + a_{k-1} p_1 + a_k p_0 &= b_k
 \end{aligned}$$

which constitute a set of $k + 1$ simultaneous equations for $p_0, p_1, \dots, p_k$.

The above equations may be arranged in vector-matrix form as

$$\begin{bmatrix} 1 & 0 & \cdot & \cdot & 0 \\ a_1 & 1 & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{k-1} & a_{k-2} & \cdot & \cdot & 0 \\ a_k & a_{k-1} & \cdot & \cdot & 1 \end{bmatrix} \begin{bmatrix} p_0 \\ p_1 \\ \cdot \\ p_{k-1} \\ p_k \end{bmatrix} = \begin{bmatrix} b_0 \\ b_1 \\ \cdot \\ b_{k-1} \\ b_k \end{bmatrix} \quad (12)$$

- The $(k + 1, k + 1)$ matrix in the above equation is known as *TOEPLITZ* matrix. The determinant of the matrix is 1. Hence it is always possible to solve for the p_i , given the numerator coefficients b_i .
- It is worth noting that although the state variables in the original first canonical form and in the alternate canonical form are identified with the outputs of the integrators, they are not the same variables. The matrix A is the same for both but not B and C . Of course, they are related by some linear transformation.

Second Companion Form

- In the first companion form, the coefficients of the denominator of the transfer function appear in one of the rows of A .
- This is another set of companion forms in which the coefficients appear in a column of the A matrix.
- Let us write the original transfer function

$$(s^k + a_1 s^{k-1} + \dots + a_k) Y(s) = (b_0 s^k + b_1 s^{k-1} + \dots + b_k) U(s)$$

or

$$s^k [Y(s) - b_0 U(s)] + s^{k-1} [a_1 Y(s) - b_1 U(s)] + \dots + [a_k Y(s) - b_k U(s)] = 0$$

- On dividing by s^k and solving for $Y(s)$, we obtain

$$Y(s) = b_0 U(s) + \frac{1}{s} [b_1 U(s) - a_1 Y(s)] + \dots + \frac{1}{s^k} [b_k U(s) - a_k Y(s)] \quad (13)$$

- Notify that the multiplier $\frac{1}{s^j}$ is the transfer function of a chain of j integrators, immediately leads to the structure as shown in the Figure below.

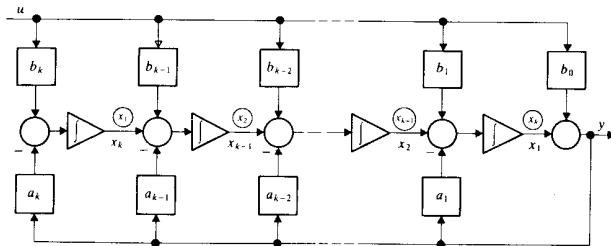


Figure: State-space realization of transfer functions in Second companion form for

$$H(s) = \frac{b_0 s^k + b_1 s^{k-1} + \dots + b_k}{s^k + a_1 s^{k-1} + \dots + a_k}$$

- The signal y is fed back to each of the integrators in the chain, and the signal u is fed forward.
- Thus the signal $b_k u - a_k y$ passes through k integrators, as required by (13), the signal $b_{k-1} u - a_{k-1} y$ passes through $k - 1$ integrators and so forth to complete the realization of (13).
- Using the state variables, the differential equations are

$$\begin{aligned}
 \dot{x}_1 &= x_2 - a_1 (x_1 + b_0 u) + b_1 u \\
 \dot{x}_2 &= x_3 - a_2 (x_1 + b_0 u) + b_2 u \\
 &\vdots \\
 &\vdots \\
 \dot{x}_{k-1} &= x_k - a_{k-1} (x_1 + b_0 u) + b_{k-1} u \\
 \dot{x}_k &= -a_k (x_1 + b_0 u) + b_k u
 \end{aligned}$$

and the output equation is

$$y = x_1 + b_0 u$$

- Thus, the matrices that describe the state space realization are given by

$$A = \begin{bmatrix} -a_1 & 1 & 0 & \cdot & \cdot & 0 \\ -a_2 & 0 & 1 & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ -a_{k-1} & 0 & 0 & \cdot & \cdot & 0 \\ -a_k & 0 & 0 & \cdot & \cdot & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} b_1 - a_1 b_0 \\ b_2 - a_2 b_0 \\ \cdot \\ \cdot \\ b_{k-1} - a_{k-1} b_0 \\ b_k - a_k b_0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & \cdot & \cdot & 0 \end{bmatrix}$$

$$D = [b_0]$$

- Just as there are two versions of the first companion form, there are also two versions of the second companion form. The second version has the structure as shown in the Figure below.

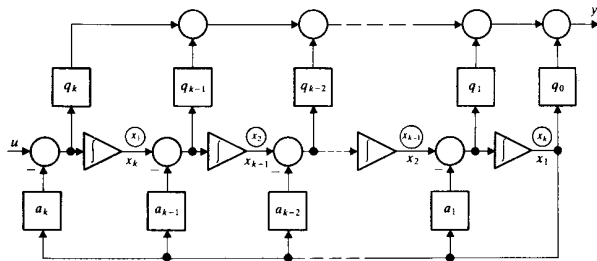


Figure: Alternative second companion form of realization of transfer function

$$H(s) = \frac{b_0 s^k + b_1 s^{k-1} + \dots + b_k}{s^k + a_1 s^{k-1} + \dots + a_k}$$

- Here also $q_1, q_2, \dots, q_k$ are not the numerator coefficients of the transfer function. They can be evaluated in terms of numerator and denominator coefficients.

Diagonal Form: Partial Fraction Expansion

- Another of the canonical forms of the realization of a transfer function is *diagonal form*, so named because of the nature of the A matrix.
- This canonical form follows directly from the partial fraction expansion of the transfer functions.
- The partial fraction expansion of the transfer function has the form

$$H(s) = b_0 + \frac{r_1}{s - s_1} + \frac{r_2}{s - s_2} + \dots + \frac{r_k}{s - s_k} \quad (14)$$

- The coefficients $r_i (i = 1, 2, \dots, k)$ are the residues of the transfer function $H(s) - b_0$ at the corresponding poles.
- In the form of (14), the transfer function consists of a direct path with gain b_0 and k first order transfer functions in parallel.

A block diagram representation of (14) is shown in Figure below as follows.

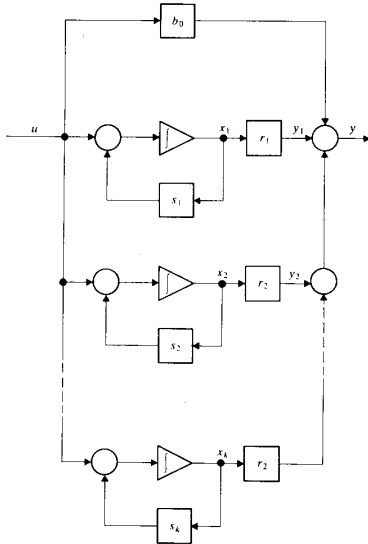


Figure: Diagonal form transfer function with distinct roots

- Identifying the outputs of the integrators with the state variables, results in the following differential equations:

$$\dot{x}_1 = s_1 x_1 + u$$

$$\dot{x}_2 = s_2 x_2 + u$$

$$\cdot \quad \cdot \quad \cdot$$

$$\cdot \quad \cdot \quad \cdot$$

$$\dot{x}_k = s_k x_k + u$$

and the observation equation

$$y = r_1 x_1 + r_2 x_2 + \dots + r_k x_k + b_0 u$$

- Hence, the matrices corresponding to this realization are

$$A = \begin{bmatrix} s_1 & 0 & \cdot & \cdot & 0 \\ 0 & s_2 & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & \cdot & \cdot & s_k \end{bmatrix}$$

$$B = \begin{bmatrix} 1 \\ 1 \\ \cdot \\ \cdot \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} r_1 & r_2 & \cdot & \cdot & r_k \end{bmatrix}$$

$$D = [b_0]$$

Note that A is a diagonal matrix.

- If the system has repeated eigenvalues, then $H(s)$ can be written as

$$H(s) = b_0 + \sum_{i=1}^r \frac{C_i}{(s - \lambda_1)^{r-i+1}} + \sum_{i=r+1}^k \frac{C_i}{s - \lambda_i}$$

$$\dot{X} = JX + BU$$

$$Y = CX + b_0 U$$

where

$$J = \begin{bmatrix} \lambda_1 & 1 & 0 & \cdot & \cdot & 0 & 0 & 0 & 0 & 0 \\ 0 & \lambda_1 & 1 & \cdot & \cdot & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda_1 & \cdot & \cdot & 0 & 0 & 0 & 0 & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 0 & 0 & 0 & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cdot & \cdot & \lambda_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{r+1} & 0 & \cdot & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{r+2} & \cdot & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdot & \lambda_{r+k} \end{bmatrix}$$

$$B = [0 \ 0 \ \cdot \ \cdot \ 1 \ 1 \ 1 \ \cdot \ \cdot \ \cdot \ 1]^T$$

$$C = [c_1 \ c_2 \ \cdot \ \cdot \ c_r \ c_{r+1} \ \cdot \ \cdot \ c_k]$$

This representation is known as the *Jordan Canonical* state model.

Thank You